# STA 243: Linear Algebra Basics

Spring 2026

# Review of Linear Algebra

Topics:

i. Span, Basis vectors, Linear Independence
ii. Range and Null space of matrices
iii. Projections
iv. Determinant and trace
v. Quadratic forms, Eigenvalues and Eigenvectors
vi. Matrix norms: Frobenius norm and operator norm
vii. Matrix calculus (we will use this for optimization)

# Vectors

- $x \in \mathbb{R}^n \to x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ called **Column vector**
- $x^\top \to$ called **Row vector**
- $x^\top y = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \sum_{i=1}^n x_i y_i \quad \longleftarrow$ a real num

  $\to$ **dot product** or **inner product**. (Here: $x, y \in \mathbb{R}^n$)

# Outer Product

For $x \in \mathbb{R}^m$ and $y \in \mathbb{R}^n$:

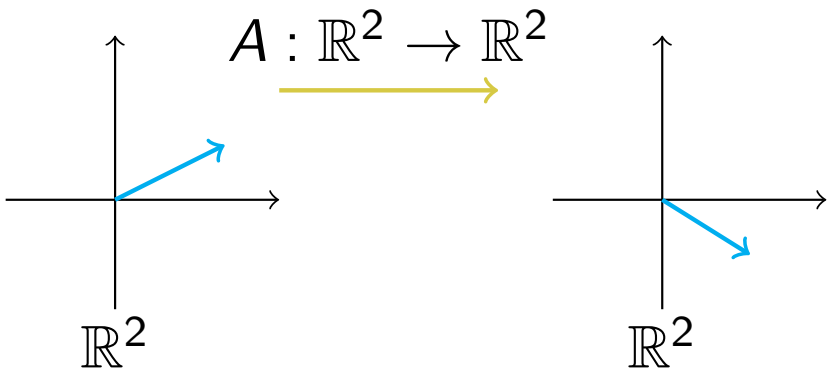
$$xy^\top = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \begin{bmatrix} y_1 & y_2 & \cdots & y_n \end{bmatrix}$$

**Outer product**:

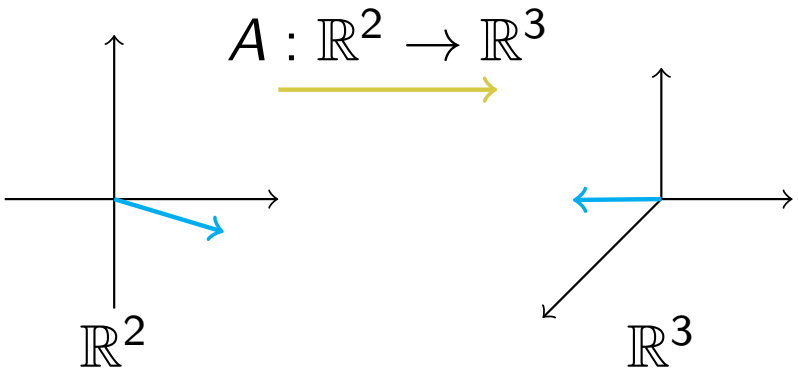
$$= \begin{bmatrix} x_1 y_1 & x_1 y_2 & \cdots & x_1 y_n \\ x_2 y_1 & x_2 y_2 & \cdots & x_2 y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_m y_1 & x_m y_2 & \cdots & x_m y_n \end{bmatrix} \in \mathbb{R}^{m \times n}$$

# What is a Matrix?

Linear Transformation. That is $A \in \mathbb{R}^{m \times n}$ is a mapping from $\mathbb{R}^n \to \mathbb{R}^m$. It takes every $n$ dimensional vector and transforms it into an $m$ dimensional vector.




- $A \in \mathbb{R}^{n \times n}$
- $A \in \mathbb{R}^{m \times n}$




- Linearity: $A(v_1 + v_2) = Av_1 + Av_2$

# Convention

$$A = \begin{bmatrix} | & | & | & & | \\ a_1 & a_2 & a_3 & \cdots & a_n \\ | & | & | & & | \end{bmatrix} \in \mathbb{R}^{m \times n}$$

Each $a_i \in \mathbb{R}^m$

# Matrix–Vector Product as Linear Combination

For $A \in \mathbb{R}^{m \times n}$ and $x \in \mathbb{R}^n$:

$$y = Ax = \begin{bmatrix} | & | & | & & | \\ a_1 & a_2 & a_3 & \cdots & a_n \\ | & | & | & & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$= x_1 \begin{bmatrix} | \\ a_1 \\ | \end{bmatrix} + x_2 \begin{bmatrix} | \\ a_2 \\ | \end{bmatrix} + \cdots + x_n \begin{bmatrix} | \\ a_n \\ | \end{bmatrix} \in \mathbb{R}^m$$

$y$ is a **linear combination of columns of $A$ matrix.**

# Matrix Multiplication

$$A \in \mathbb{R}^{m\times n}, \quad B \in \mathbb{R}^{n\times p}, \quad C = AB$$

$$C_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$$

$\boxed{\text{In general } AB \neq BA}$

**Viewpoint 1: Inner product**

$$\begin{bmatrix} - & a_1^\top & - \\ - & a_2^\top & - \\ & \vdots & \\ - & a_m^\top & - \end{bmatrix} \begin{bmatrix} | & | & & | \\ b_1 & b_2 & \cdots & b_p \\ | & | & & | \end{bmatrix} = \begin{bmatrix} a_1^\top b_1 & a_1^\top b_2 & \cdots & a_1^\top b_p \\ a_2^\top b_1 & a_2^\top b_2 & \cdots & a_2^\top b_p \\ \vdots & \vdots & \ddots & \vdots \\ a_m^\top b_1 & a_m^\top b_2 & \cdots & a_m^\top b_p \end{bmatrix}$$

# Matrix Multiplication — Viewpoint 2

**Viewpoint 2:**

$$C = AB = \begin{bmatrix} | & | & & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} - & b_1^\top & - \\ - & b_2^\top & - \\ & \vdots & \\ - & b_n^\top & - \end{bmatrix}$$

$$= \sum_{i=1}^{n} a_i b_i^\top$$

$$= a_1 b_1^\top + a_2 b_2^\top + \cdots + a_n b_n^\top$$

(Sum of outer products — each $a_i b_i^\top$ is an $m \times p$ matrix)

# Transpose

$$(A^\top)_{ij} = A_{ji}$$

- $(A^\top)^\top = A$
- $(AB)^\top = B^\top A^\top$
- $(A+B)^\top = A^\top + B^\top$ $\Big\}$

If for $A \in \mathbb{R}^{n\times n}$ square, $A = A^\top$, then we say $A$ is a <u>symmetric matrix</u>.

# Trace

For $A \in \mathbb{R}^{n \times n}$:

$$\text{trace}(A) = \sum_{i=1}^{n} A_{ii}$$

[Usually has an "energy" interpretation]

- $\text{trace}(A) = \text{trace}(A^\top)$
- $\text{trace}(A + B) = \text{trace}(A) + \text{trace}(B)$
- Trace is cyclic:

$$\text{trace}(ABC) = \text{trace}(CAB) = \text{trace}(BCA)$$

[Try to prove it!]

# Inverse

The inverse of a square matrix $A \in \mathbb{R}^{n \times n}$ is a matrix $A^{-1}$ which satisfies:

$$A^{-1}A = I = AA^{-1}$$

where

$$I = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} \in \mathbb{R}^{n \times n}$$

is the identity matrix.

# Inverse (cont.)

- Not all matrices have inverse.
  - Non-square matrices do not have inverse by definition.
  - Even for some square matrices, inverse may not exist!
- $(A^{-1})^{-1} = A$
- $(AB)^{-1} = B^{-1}A^{-1}$

# Linear Independence

$x_1, x_2, \ldots, x_n \in \mathbb{R}^m$ is said to be **linearly independent** if no vector can be written as linear combination of the remaining vectors.

Dependent: $x_n = \sum_{i=1}^{n-1} \alpha_i x_i \quad \longrightarrow$ for some $\alpha_i$'s not all 0

e.g.: $x_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 4 \\ 1 \\ 5 \end{bmatrix}, \quad x_3 = \begin{bmatrix} 2 \\ -3 \\ -1 \end{bmatrix}$

$$x_3 = -2x_1 + x_2$$

# Column Rank and Row Rank

- **Column rank** of $A \in \mathbb{R}^{m \times n}$:
  $\rightarrow$ size of the largest subset of columns of $A$ that are linearly independent
- **Row rank**: Same for rows
- For a matrix, row rank = column rank (called just "rank")

# Rank and Invertibility

- $A \in \mathbb{R}^{m \times n}, \quad \text{rank}(A) \leq \min(m, n)$
- If $\text{rank}(A) = \min(m, n)$ then $A$ is **full rank**
- For a square matrix to be invertible, it must have full rank.
  $\rightarrow$ For $A \in \mathbb{R}^{n \times n}$, if $\text{rank}(A) = n$ then it is invertible

# Range and Null Space

**Span:**

$$\text{span}(\{x_1, x_2, \ldots, x_n\}) = \left\{ v : v = \sum_{i=1}^{n} \alpha_i x_i, \quad \alpha_i \in \mathbb{R} \right\}$$

$A \in \mathbb{R}^{m \times n} \rightarrow$ what is Range$(A)$?

$$\begin{aligned} R(A) &= \{v \in \mathbb{R}^m : v = Ax, \quad x \in \mathbb{R}^n\} \\ &= \left\{ v \in \mathbb{R}^m : v = \sum_{i=1}^{n} x_i a_i, \quad x_i \in \mathbb{R} \right\} \end{aligned}$$

where $a_i \rightarrow$ columns of $A$ (vectors in $\mathbb{R}^m$)

**Note:** $R(A^\top)$ **composes of vectors in** $\mathbb{R}^n$ **s.t. ...**

# Null Space

$$N(A) = \{x \in \mathbb{R}^n : Ax = 0\}$$

$$Ax = x_1 \begin{bmatrix} | \\ a_1 \\ | \end{bmatrix} + x_2 \begin{bmatrix} | \\ a_2 \\ | \end{bmatrix} + \cdots + x_n \begin{bmatrix} | \\ a_n \\ | \end{bmatrix} = 0$$

$$\begin{bmatrix} | \\ a_n \\ | \end{bmatrix} = -\frac{x_1}{x_n} \begin{bmatrix} | \\ a_1 \\ | \end{bmatrix} - \frac{x_2}{x_n} \begin{bmatrix} | \\ a_2 \\ | \end{bmatrix} - \cdots - \frac{x_{n-1}}{x_n} \begin{bmatrix} | \\ a_{n-1} \\ | \end{bmatrix}$$

$\Leftrightarrow$ linear independence of columns of $A$

# Fundamental Subspaces

Given $A \in \mathbb{R}^{m \times n}$,

- $R(A)$ and $N(A^\top)$ are orthogonal complements in $\mathbb{R}^m$
- $R(A^\top)$ and $N(A)$ are orthogonal complements in $\mathbb{R}^n$

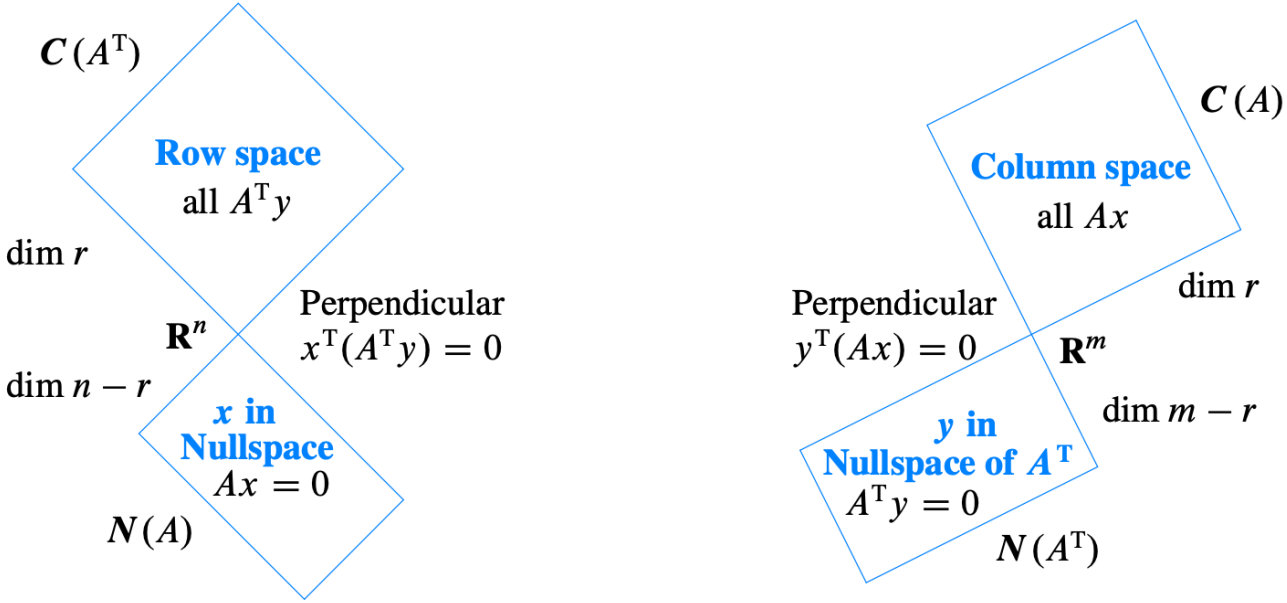



Figure 1: Dimensions and orthogonality for any $m$ by $n$ matrix $A$ of rank $r$.

# What is Least-Square Regression?

Given $a_1, \ldots, a_n \in \mathbb{R}^d, \quad b_1, \ldots, b_n \in \mathbb{R}$
Assume $b_i = a_i^\top x^* + \varepsilon$

With $A \in \mathbb{R}^{n \times d}$ and $n > d, \quad b \in \mathbb{R}^n$

$$\hat{x} = (A^\top A)^{-1} A^\top b$$

$$\hat{b} = A(A^\top A)^{-1} A^\top b \;\rightarrow\; \hat{b} \in R(A)$$

$\hat{b} \rightarrow$ closest vector in $R(A)$ to $b$

**Note:** Related to pseudo-inverse $A^\dagger$

# Basis Vectors

A set of vectors $\{v_1, v_2, \ldots, v_k\}$ in $\mathbb{R}^m$ is called a **basis** for a subspace $S \subseteq \mathbb{R}^m$ if:

1. $v_1, \ldots, v_k$ are linearly independent
2. $\text{span}(\{v_1, \ldots, v_k\}) = S$

- The number of vectors in any basis for $S$ is always the same $\rightarrow$ called the **dimension** of $S$.
- The standard basis for $\mathbb{R}^n$:

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \ldots, \quad e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

- Any $x \in \mathbb{R}^n$ can be written as $x = x_1 e_1 + x_2 e_2 + \cdots + x_n e_n$

# Basis Vectors (cont.)

- A basis is **not unique**! Many different sets of vectors can serve as a basis for the same subspace.
- An **orthonormal basis** $\{u_1, \ldots, u_k\}$ satisfies:

$$u_i^\top u_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

- Orthonormal bases are especially convenient for computation.
- The columns of $A \in \mathbb{R}^{m \times n}$ form a basis for $R(A)$ if and only if $\text{rank}(A) = n$.

# Projections

**Orthogonal projection** of a vector $b$ onto a subspace $S$:
Find the vector $\hat{b} \in S$ that is closest to $b$.

**Projection onto a vector:** Given $a \in \mathbb{R}^m$, the projection of $b$ onto $a$ is:

$$\text{proj}_a(b) = \frac{a^\top b}{a^\top a}\, a$$

Key property: the residual $b - \text{proj}_a(b)$ is orthogonal to $a$:

$$a^\top \left( b - \frac{a^\top b}{a^\top a} a \right) = 0$$

# Projections onto a Subspace

**Projection onto $R(A)$:** For $A \in \mathbb{R}^{m \times n}$ with full column rank, the projection of $b \in \mathbb{R}^m$ onto $R(A)$ is:

$$\hat{b} = A(A^\top A)^{-1} A^\top b$$

The matrix $P = A(A^\top A)^{-1} A^\top$ is the **projection matrix**.

Properties of projection matrices:

- $P^2 = P$ (idempotent — projecting twice is the same as projecting once)
- $P^\top = P$ (symmetric)
- $b - Pb \perp R(A)$ (residual is orthogonal to the subspace)
- If $A$ has orthonormal columns ($A^\top A = I$), then $P = AA^\top$

# Determinant

For $A \in \mathbb{R}^{n \times n}$, the **determinant** $\det(A)$ is a scalar that captures how the matrix scales volumes.

For $2 \times 2$:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \det(A) = ad - bc$$

For $3 \times 3$: expand along a row or column (cofactor expansion).

**Geometric interpretation:** $|\det(A)|$ gives the factor by which $A$ scales volumes. If $\det(A) < 0$, the transformation reverses orientation.

# Determinant — Properties

- $\det(I) = 1$
- $\det(A^\top) = \det(A)$
- $\det(AB) = \det(A)\det(B)$
- $\det(A^{-1}) = \frac{1}{\det(A)}$
- $\det(cA) = c^n \det(A) \quad \text{for } A \in \mathbb{R}^{n\times n}$
- $A$ is invertible $\iff \det(A) \neq 0$
- If $A$ has eigenvalues $\lambda_1, \ldots, \lambda_n$, then $\det(A) = \prod_{i=1}^n \lambda_i$

Note: $\text{trace}(A) = \sum_{i=1}^n \lambda_i$ and $\det(A) = \prod_{i=1}^n \lambda_i$.

# Eigenvalues and Eigenvectors

For $A \in \mathbb{R}^{n \times n}$, a non-zero vector $v \in \mathbb{R}^n$ is an **eigenvector** of $A$ if:

$$Av = \lambda v$$

for some scalar $\lambda \in \mathbb{R}$ (or $\mathbb{C}$), called the corresponding **eigenvalue**.

**Interpretation:** The matrix $A$ acts on $v$ by simply scaling it – the direction does not change (only the magnitude).

To find eigenvalues, solve:

$$\det(A - \lambda I) = 0 \quad \text{(characteristic polynomial)}$$

Then for each $\lambda_i$, solve $(A - \lambda_i I)v = 0$ to find the eigenvectors.

# Eigendecomposition

If $A \in \mathbb{R}^{n \times n}$ is **symmetric** ($A = A^\top$), then:

- All eigenvalues are **real**
- Eigenvectors corresponding to distinct eigenvalues are **orthogonal**
- $A$ has the **spectral decomposition**:

$$A = U \Lambda U^\top = \sum_{i=1}^{n} \lambda_i u_i u_i^\top$$

where $U = [u_1 \ u_2 \ \cdots \ u_n]$ is orthogonal ($U^\top U = I$) and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$.

This decomposition is extremely useful throughout computational statistics!

# Quadratic Forms

For a symmetric matrix $A \in \mathbb{R}^{n \times n}$, the **quadratic form** associated with $A$ is:

$$f_A(x) = x^\top A x = \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} x_i x_j$$

- $A$ is **positive semidefinite** (PSD), written $A \succeq 0$, if:

$$x^\top A x \geq 0 \quad \text{for all } x \in \mathbb{R}^n$$

- $A$ is **positive definite** (PD), written $A \succ 0$, if:

$$x^\top A x > 0 \quad \text{for all } x \neq 0$$

- $A \succeq 0 \iff$ all eigenvalues $\lambda_i \geq 0$
- $A \succ 0 \iff$ all eigenvalues $\lambda_i > 0$

# Quadratic Forms — Examples

- For any matrix $B \in \mathbb{R}^{m \times n}$, the matrix $B^\top B \in \mathbb{R}^{n \times n}$ is always PSD:
$$x^\top (B^\top B) x = (Bx)^\top (Bx) = \|Bx\|^2 \geq 0$$
- The covariance matrix $\Sigma = \mathbb{E}[(X - \mu)(X - \mu)^\top]$ is always PSD.
- Quadratic forms appear everywhere in statistics:
  - Least squares: $\|Ax - b\|^2 = x^\top A^\top A x - 2b^\top A x + b^\top b$
  - Gaussian density: $\exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1} (x - \mu)\right)$
  - Mahalanobis distance: $(x - \mu)^\top \Sigma^{-1} (x - \mu)$

# Matrix Norms: Frobenius Norm

For $A \in \mathbb{R}^{m \times n}$, the **Frobenius norm** is:

$$\|A\|_F = \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij}^2} = \sqrt{\text{trace}(A^\top A)}$$

- Think of it as "flattening the matrix into a vector and taking the Euclidean norm."
- In terms of singular values: $\|A\|_F = \sqrt{\sum_i \sigma_i^2}$
- In terms of eigenvalues (symmetric $A$): $\|A\|_F = \sqrt{\sum_i \lambda_i^2}$

Properties:

- $\|A\|_F \geq 0$, with equality iff $A = 0$
- $\|cA\|_F = |c| \, \|A\|_F$
- $\|A + B\|_F \leq \|A\|_F + \|B\|_F$ (triangle inequality)

# Matrix Norms: Operator Norm (Spectral Norm)

The **operator norm** (or spectral norm) of $A \in \mathbb{R}^{m \times n}$ is:

$$\|A\|_{\text{op}} = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \max_{\|x\|=1} \|Ax\|$$

- It measures the maximum "stretching" that $A$ can do to any unit vector.
- $\|A\|_{\text{op}} = \sigma_{\max}(A)$ (largest singular value of $A$)
- For symmetric $A$: $\|A\|_{\text{op}} = \max_i |\lambda_i|$ (largest absolute eigenvalue)

**Relationship:**

$$\|A\|_{\text{op}} \leq \|A\|_F \leq \sqrt{r}\, \|A\|_{\text{op}}$$

where $r = \text{rank}(A)$.

# Matrix Calculus — Gradients

We will use matrix calculus for optimization. Key results:

For $x \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and symmetric $S \in \mathbb{R}^{n \times n}$:

- $\nabla_x (a^\top x) = a$
- $\nabla_x (x^\top a) = a$
- $\nabla_x (x^\top S x) = 2Sx \quad (S \text{ symmetric})$
- $\nabla_x \|Ax - b\|^2 = 2A^\top (Ax - b)$

**Example:** Least squares minimization of $f(x) = \|Ax - b\|^2$:

$$\nabla_x f = 2A^\top (Ax - b) = 0 \implies A^\top A x = A^\top b \implies \hat{x} = (A^\top A)^{-1} A^\top b$$

# Matrix Calculus — Useful Identities

Derivatives with respect to matrices: for $A \in \mathbb{R}^{m \times n}$

- $\frac{\partial}{\partial A}\text{trace}(AB) = B^\top$
- $\frac{\partial}{\partial A}\text{trace}(A^\top B) = B$
- $\frac{\partial}{\partial A}\text{trace}(A^\top A) = 2A$

For invertible square $A \in \mathbb{R}^{n \times n}$:

- $\frac{\partial}{\partial A} \log \det(A) = (A^{-1})^\top = A^{-\top}$
- $\frac{\partial}{\partial A} \det(A) = \det(A)\, A^{-\top}$

These identities are essential for deriving estimators in computational statistics (e.g., maximum likelihood for Gaussian models).

# Additional Linear Algebra

# Variational Characterization of Eigenvalues

Let $A \in \mathbb{R}^{n \times n}$ be symmetric with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$.

**Theorem (Courant–Fischer).** For $k \in \{1, \ldots, n\}$:

$$\lambda_k(A) = \max_{\substack{V \subseteq \mathbb{R}^n \\ \dim(V)=k}} \min_{\substack{x \in V \\ \|x\|=1}} \langle x, Ax \rangle = \min_{\substack{V \subseteq \mathbb{R}^n \\ \dim(V)=n-k+1}} \max_{\substack{x \in V \\ \|x\|=1}} \langle x, Ax \rangle$$

**Special cases:**

- $k = 1$: $\lambda_1(A) = \max_{\|x\|=1} \langle x, Ax \rangle$
- $k = n$: $\lambda_n(A) = \min_{\|x\|=1} \langle x, Ax \rangle$

# Best Rank-$k$ Approximation

**Problem:** Given $M \in \mathbb{R}^{m \times n}$, find the closest rank-$k$ matrix.

$$\min_{Z} \ \|M - Z\|_F^2 \quad \text{s.t.} \quad \mathrm{rank}(Z) \leq k$$

**Theorem (Eckart–Young).** The solution is:

$$Z^* = \sum_{i=1}^{k} \sigma_i \, u_i v_i^\top = MP_k$$

where $P_k$ projects onto $\mathrm{span}(v_1, \ldots, v_k)$.

**Residual error:**

$$\|M - Z^*\|_F^2 = \sum_{i=k+1}^{\min(m,n)} \sigma_i(M)^2$$

# Eckart–Young: Proof Idea

**Key identity.** For any unit vector $v \in \mathbb{R}^n$:

$$\|M - Mvv^\top\|_F^2 = \|M\|_F^2 - \|Mv\|_2^2$$

So minimizing the residual $\Leftrightarrow$ maximizing $\|Mv\|_2^2$.

By the variational characterization of $\sigma_1$:

$$\min_{\|v\|=1} \|M - Mvv^\top\|_F^2 = \|M\|_F^2 - \sigma_1(M)^2 = \sum_{i=2}^{\min(m,n)} \sigma_i(M)^2$$

**Generalization:** Replace $vv^\top$ with a rank-$k$ projector $P$:

$$\min_{\substack{P=P^\top,\, P^2=P \\ \mathrm{rank}(P)\le k}} \|M - MP\|_F^2 = \sum_{i=k+1}^{\min(m,n)} \sigma_i(M)^2$$

Minimizer: $P^* =$ projector onto $\mathrm{span}(v_1, \ldots, v_k)$. $\square$

# Weyl's Inequality: Eigenvalue Perturbation

How much can eigenvalues change under perturbation?

**Theorem (Weyl).** For $A, B \in \mathbb{R}^{n \times n}$ symmetric:

$$|\lambda_k(A+B) - \lambda_k(A)| \leq \|B\|_{\mathrm{op}} \quad \text{for all } k = 1, \ldots, n$$

**Proof sketch (upper bound):**

$$\begin{aligned}
\lambda_k(A+B) &= \max_{\dim(V)=k} \min_{\substack{x \in V \\ \|x\|=1}} \big(\langle x, Ax\rangle + \langle x, Bx\rangle\big) \\
&\leq \max_{\dim(V)=k} \min_{\substack{x \in V \\ \|x\|=1}} \langle x, Ax\rangle + \lambda_1(B) \\
&= \lambda_k(A) + \|B\|_{\mathrm{op}}
\end{aligned}$$

Similarly: $\lambda_k(A+B) \geq \lambda_k(A) + \lambda_n(B) \geq \lambda_k(A) - \|B\|_{\mathrm{op}}$. $\square$

# Consequences of Weyl's Inequality

**Singular value version.** For $M, N \in \mathbb{R}^{m \times n}$:

$$|\sigma_k(M+N) - \sigma_k(M)| \leq \|N\|_{\mathrm{op}} = \sigma_1(N)$$

**Operator norm is a norm.** Triangle inequality:

$$\big| \|A+B\|_{\mathrm{op}} - \|A\|_{\mathrm{op}} \big| \leq \|B\|_{\mathrm{op}}$$

**Interpretation:** If $\hat{A} = A + E$ is a noisy observation of $A$, then:

$$|\lambda_k(\hat{A}) - \lambda_k(A)| \leq \|E\|_{\mathrm{op}}$$

$\Rightarrow$ Eigenvalues are **Lipschitz-stable** under perturbation.
$\rightsquigarrow$ This is the starting point for analyzing spectral methods on noisy data

# Coming up

We'll start talking about randomized linear algebra next time!